MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG2DSCIAM100 Intelligent Automation Techniques

Duration: 1.5 hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	Which of the following best defines Artificial Intelligence (AI)? a) The ability of machines to perform tasks without human intervention b) A branch of computer science that aims to create intelligent machines c) A software program that mimics human behavior d) A set of rules followed by a computer	K	1
2	What is a key characteristic of neural networks in machine learning? a) They are rule-based systems b) They process information in layers c) They do not require large datasets d) They work only with structured data	U	1
3	Which of the following is NOT a type of machine learning system? a) Supervised Learning b) Reinforcement Learning c) Predictive Learning d) Unsupervised Learning	U	1
4	Which library is commonly used to extract data from a MySQL database in Python? a) NumPy b) Matplotlib c) Pandas d) TensorFlow	K	2
5	Which Python library is commonly used to extract data from a database? a) pandas b) matplotlib c) numpy d) seaborn	U	2
6	Which of the following is NOT a valid way to run Python code? a) Jupyter Notebook b) Pycharm c) Microsoft Excel d) IDLE	K	2
7	In Gradient Descent, what does the learning rate control? a) The number of neurons b) The size of the step taken towards the minimum c) The number of features in the dataset d) The type of activation function used	U	2
8	In reinforcement learning, the system learns by: a) Finding hidden patterns in unlabeled data b) Comparing its predictions with true labels c) Receiving rewards or penalties for actions taken d) Assigning data points to clusters	K	3
9	Which of the following is a type of regression analysis? a) Hierarchical clustering b) Decision trees c) Linear regression d) SVM	U	3
10	Which of the following is an example of supervised learning? a) K-Means Clustering b) Principal Component Analysis c) K-Nearest Neighbors (KNN) d) Hierarchical Clustering	U	3

1 * 10 = 10

Part B

Short Answer Questions

Answer any **4** Questions

Each question carries 5 marks

11	Mention two major challenges in implementing machine learning models.	U	1
12	What are variables in Python? Give an example.	A	2
13	What is the primary purpose of the gradient descent algorithm?	U	2
14	Explain the working principle of Hierarchical Clustering with an example.	A	3
15	What is the purpose of regression analysis in machine learning?	U	3
16	What is the primary difference between supervised and unsupervised machine learning?	U	3

4 * 5 = 20

Part C

Essay Questions

Answer any **2** Questions

Each question carries 10 marks

17	Explain the different types of machine learning systems with examples.	U	1
18	Discuss the applications of machine learning in medical treatment and business, highlighting their impact.	U	1
19	Describe Python's control flow statements (if-elif-else, loops) with syntax and examples.	A	2
20	Describe the working of K-Nearest Neighbors (KNN) and its applications in machine learning.	U	3

2 * 10 = 20